ViBridge-VQA: Bridging Global and Local Visual Tokens for Parameter-Efficient Vietnamese Visual Question Answering

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Abstract. Adapting vision–language models (VLMs) such as Vintern-1B to a new Vietnamese Visual Question Answering (VQA) benchmark typically requires fine-tuning on many high-resolution image tiles or a new architecture. We propose **ViBridge-VQA**, which improves Vintern-1B without updating its pre-trained weights. From a single image view, the frozen decoder receives *global* tokens from the `[CLS]` vector through a small trainable bridge, and *local* tokens from Vintern-1B's pre-aligned projector, pooled to a budget chosen at the knee of the accuracy–cost Pareto front. On AutoViVQA, ViBridge-VQA trains 1.35% of the parameters, uses 36 times fewer visual tokens than official fine-tuning, and achieves 54.6 F1 on an image-disjoint split; a lightweight decoder LoRA raises F1 to 58.2. Reusing the pre-trained projector outperforms the learned connectors evaluated here, and local tokens also improve results on two other Vietnamese VQA datasets.

Keywords: Visual Question Answering · Vietnamese · Vision–Language Models · Visual Token Compression · Parameter-Efficient Adaptation

1 Introduction

Visual Question Answering (VQA) asks a model to answer a natural-language question about an image. Large vision–language models (VLMs) such as BLIP-2 [13], Flamingo [1] and LLaVA [20] generate free-form answers by connecting a pre-trained vision encoder to a pre-trained language model. For Vietnamese, the open Vintern-1B [8] connects InternViT-300M, a distilled InternVL encoder [7], to a small Qwen decoder (Qwen2.5-0.5B [27] in the v3.5 release we use), and benchmarks such as ViVQA [30] and the reasoning-oriented AutoViVQA [22] measure progress in this low-resource setting.

Adapting a VLM to a new benchmark remains expensive. The fine-tuned Vintern-1B baseline in AutoViVQA adds LoRA adapters and splits each test

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image into six high-resolution tiles plus a thumbnail, yielding 1,792 visual tokens per image. ViMoE-VQA [21] instead introduces a task-specific mixture-of-experts fusion module. A cheaper alternative freezes the VLM and trains a small connector. A *global-only bridge* maps the global image vector to a few tokens, but loses small objects, printed text, and spatial relations. A *learned patch compressor*, such as a Q-Former, retains these details but must learn a new mapping to the decoder from limited Vietnamese data.

Vintern-1B offers a third option: its projector already maps image regions into decoder input space, so this alignment need not be learned again. It also discards the encoder's global `[CLS]` vector, which summarises the whole scene. We therefore ask whether reusing this alignment is more effective than learning a new connector with limited labelled data, and how many global and local tokens the frozen decoder needs.

We propose **ViBridge-VQA**, which feeds the frozen decoder of Vintern-1B two complementary kinds of visual tokens from a single image view. *Local* tokens are obtained from the frozen projector, average-pooled to a selected budget, and inserted into the image slot, preserving the representation the decoder was trained to read. *Global* tokens, produced from `[CLS]` by a small trainable bridge, provide scene-level information that Vintern-1B otherwise ignores. By default, only the bridge is trainable, leaving Vintern-1B unchanged and reusable across tasks. Because tokens add inference cost, we treat their count as a budget and select the knee of the accuracy–cost Pareto front. Without updating Vintern-1B, ViBridge-VQA achieves an F1 close to that reported for the fine-tuned model while using 36 times fewer visual tokens. Our contributions are:

- **ViBridge-VQA**, a global–local bridge that combines trainable global tokens with local tokens from the frozen projector of the VLM, optionally with a lightweight decoder LoRA.
- An explicit **accuracy–cost budget** for the number of visual tokens, selected on validation as the knee of the Pareto front.
- An **analysis of the source of the gains**: in our setup, reusing the pre-aligned projector is more data-efficient than learning a new connector and outperforms learned connectors with up to 9.4× more parameters (by 6.0 F1 over a Q-Former at a similar token count). Local tokens contribute more than global tokens and also help on two other Vietnamese VQA datasets.

2 Related Work

Vietnamese VQA. Vietnamese VQA datasets include ViVQA [30], synthetic ViCLEVR [32], book-cover ViOCRVQA [26], and scene-text ViTextVQA [24]. AutoViVQA [22] contains 37,077 questions over 19,411 COCO images [18], generated by large language models, filtered by validators, and labelled with reasoning types. On the modelling side, BARTPhoBEiT [31] and AViVQA-TranConI [23] combine pre-trained visual and Vietnamese language representations, Vintern-1B [8] brings an instruction-tuned VLM to Vietnamese, and ViMoE-VQA [21] routes multimodal tokens to sparse experts for generative VQA on AutoViVQA.

AutoViVQA and ViMoE-VQA are prior work from our group; we use the former but no component of the latter. The method, image-disjoint split, and all results here are new. The approaches above train a new model or fine-tune a VLM on multi-tile inputs, whereas we keep Vintern-1B frozen and use a single image view.

Connectors. Connectors between frozen vision encoders and language models differ in the mapping they learn. Frozen [33] learns a visual prefix from a global image representation. BLIP-2's Q-Former [13] compresses patch features with learned queries, but is a 188M-parameter module pre-trained on millions of image–text pairs. LLaVA [20,19] preserves locality by projecting every patch token, at the cost of hundreds of decoder tokens. Honeybee [4] proposes locality-preserving abstractors, while InternVL [6,7], Vintern-1B's basis, merges neighbouring patches before projection. All learn the patch-to-decoder mapping during large-scale pre-training. ViBridge-VQA instead reuses that mapping and trains only a 12.9M-parameter global bridge on roughly 26k task questions.

Visual token budget. The number of visual tokens largely determines decoder cost. Inference-optimal scaling of VLMs [14] shows that very few visual tokens can suffice for visual reasoning, and Matryoshka multimodal models [3] show with nested pooled budgets that the required budget depends on the task. Other work reduces the tokens directly: DeCo [36] replaces the Q-Former with adaptive average pooling of the patch tokens, TokenPacker [15] learns a compact projector, LLaVA-PruMerge [29] and FastV [5] prune or merge the tokens of an existing model, and VisionZip [35] keeps those that receive the most attention from `[CLS]`. Like DeCo, we use average pooling, but apply it to the output of an already pre-trained projector while keeping the entire VLM frozen. We select the budget using explicit criteria: the knee of the Pareto front [28] and TOPSIS [12]. Parameter-efficient methods such as adapters [10], prefix tuning [16] and LoRA [11] reduce the cost of adaptation from the other side, by training a few additional weights in the decoder. We use decoder LoRA as a baseline and as a complement.

3 Proposed Method

We formulate VQA as conditional answer generation: given image I and Vietnamese question q, the model generates a short free-form answer $y = (y_1, \ldots, y_T)$. ViBridge-VQA freezes all pre-trained weights of Vintern-1B [8] and changes only the decoder's visual input.

3.1 Architecture

As shown in Fig. 1, ViBridge-VQA retains Vintern-1B's three components: the InternViT-300M vision encoder (24 layers, width 1024), two-layer MLP projector P, and Qwen2.5-0.5B-Instruct decoder (24 layers, width 896). Each image is resized to one 336×336 view (24×24 patches of 14×14 pixels). InternViT, pre-trained at 448×448, bicubically interpolates its position embeddings, roughly

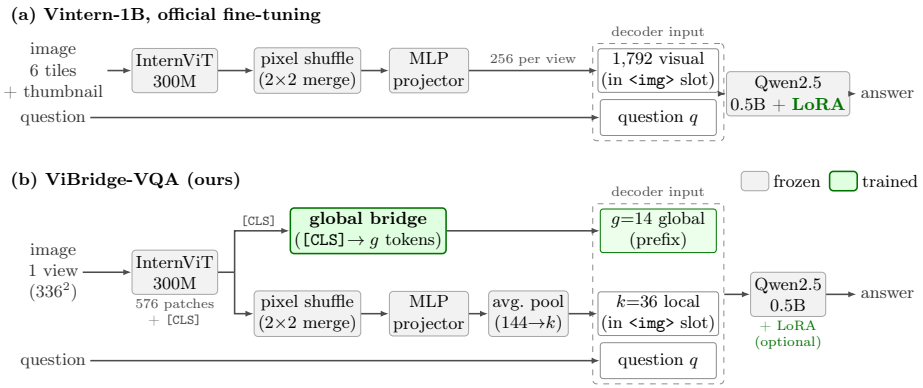


Fig. 1: Overview. (a) Official fine-tuning of Vintern-1B, with six tiles plus a thumbnail for every AutoViVQA test image. (b) ViBridge-VQA on one view: k local tokens, average-pooled from the 144 region tokens of the frozen projector, fill the image slot, and g global tokens from `[CLS]` precede the prompt. The selected budget is $g = 14$, $k = 36$ (50 tokens). The decoder stays frozen; a LoRA is optional.

halving encoder cost (191 rather than 362 GMACs). The encoder returns patch vectors and a global `[CLS]` vector $v \in \mathbb{R}^{1024}$. Vintern-1B uses only the patch vectors: pixel shuffle [6] merges each 2×2 block into a 4096-dimensional vector, which the pre-trained P maps into decoder input space. This yields a 12×12 grid of 144 region tokens. ViBridge-VQA uses both encoder outputs.

3.2 Global Tokens

The `[CLS]` vector summarises the entire scene, but Vintern-1B never passes it to the decoder. A small trainable bridge f_ϕ expands it into g tokens with two linear heads:

$$f_\phi(v) = \left[\, W_a v + b_a;\ \mathrm{reshape}_{(g-1)\times 896}(W_e v + b_e) \,\right] \in \mathbb{R}^{g \times 896}, \tag{1}$$

where $W_a \in \mathbb{R}^{896\times 1024}$ and $W_e \in \mathbb{R}^{(g-1)896\times 1024}$. The first head produces an *anchor* token, and the second produces $g-1$ complementary tokens that represent global content along different learned directions. The bridge has $1025 \cdot 896 \cdot g$ parameters (12,857,600 for $g = 14$), and its tokens are placed before the prompt. When the bridge alone is varied from 4 to 18 tokens, F1 peaks at $g = 14$ (Section 4.5), the value used in ViBridge-VQA. Our Global-only baseline uses $g = 8$, matching the eight output tokens of Tile-Attention and the light Q-Former (Section 4.4).

3.3 Local Tokens

To provide region-level detail, ViBridge-VQA passes local tokens directly from the 144 region tokens of P. We average-pool the 12×12 grid to an $s \times s$ grid, where

$k = s^2$ and $s \in \{1, 3, 6, 12\}$: $k = 144$ keeps every region token, whereas $k = 36$ averages each 2×2 block. These tokens replace k placeholders in Vintern-1B's image slot, `<img>` $\cdots$ `</img>`. They require no parameters and remain in the representation the decoder was trained to read.

3.4 Training

By default, only bridge parameters ϕ are trained; the encoder, projector, and decoder remain frozen. We minimise answer-token cross-entropy:

$$\mathcal{L} = -\sum_{t=1}^{T} \log p\big(y_t \mid y_{<t},\, f_\phi(v),\, \mathrm{pool}_s(P(x)),\, q\big), \tag{2}$$

where x denotes merged patch vectors. Optionally, a rank-16 LoRA [11] on every decoder layer's query, key, value, and output projections is trained jointly with the bridge, adding 2,162,688 parameters. Percentages are relative to all model parameters (the frozen 0.94B plus trained parameters). With $g = 14$, the bridge trains 12.86M parameters (1.35%); adding LoRA raises this to 15.02M (1.57%).

3.5 Budget Selection

Decoder cost grows with the visual-token count $g + k$, so we treat (g, k) as a budget. In our experiments, F1 increases with each added local token, placing all such configurations on the accuracy–token Pareto front. We select its *knee* [28], where returns diminish. With validation F1 a and token cost c, both normalised to $[0, 1]$, the knee maximises $\hat{a} - \hat{c}$. Equivalently, it is furthest above the line joining the cheapest and most accurate configurations. We confirm the choice with equal-weight TOPSIS [12] over accuracy (F1, CIDEr, METEOR) and cost (tokens, evaluation time, trained parameters, seed variance). Both use validation data only; code for all rankings is released with the paper.

4 Experiments

4.1 Dataset and Metrics

Dataset. We use AutoViVQA [22], which is built on MS COCO images [18]. Each question has five short free-form answers, a reasoning type, and a reasoning level. After removing 273 duplicated (image, question) pairs and 97 questions without a reasoning label, we split the remaining questions by image into 70/15/15 subsets (seed 42), ensuring that no test image appears in training (Table 1). This split is stricter than a question-level split, on which Global-only scores roughly 0.5 F1 higher.

Table 1: Our AutoViVQA split.

	Train	Val	Test	Total
Images	13,579	2,909	2,910	19,398
Questions	25,776	5,463	5,468	36,707
Answers per question				5
Mean answer length (words)				4.3
Reasoning types / levels				8 / 5

Metrics. Following AutoViVQA and ViMoE-VQA, we report Accuracy (Acc), word-level Precision (Prec), Recall (Rec), F1, BLEU [25], ROUGE-L (R-L) [17], METEOR (MET) [2], and CIDEr [34] against the five references. Precision, Recall, and F1 are computed from the word sets of each answer and its closest reference, then averaged over questions. Accuracy is the proportion of answers identical to at least one reference; BLEU is corpus-level BLEU-4.

4.2 Setup

Implementation details. The encoder, projector, and decoder are taken from Vintern-1B-v3.5 [8] and remain frozen. We train with AdamW (learning rate 2×10^{-4}; effective batch size 8) on the first reference answer. Bridges are trained for two epochs, after which validation loss and CIDEr no longer improve. Models with a decoder LoRA are trained for one epoch, with the bridge (if any) and LoRA learned jointly from scratch. We also report Global-only + LoRA after three epochs, its best setting; Table 7 and Fig. 3 use this version. We use greedy decoding with at most 50 new tokens, and every model fits on a single 16 GB NVIDIA T4. Each configuration is trained with three seeds (42, 123, 3407), except Global-only, which uses four (including 2026); tables report mean ± std.

Baselines. Table 2 compares ViBridge-VQA with two groups of models: (1) models fine-tuned on AutoViVQA: ViT5_ViT, BARTPhoBEiT [31], Vintern-1B and ViMoE-VQA [21], and (2) zero-shot models: Vintern-1B, LLaMA 3.2, Gemini 2.0 and 2.5 Flash, and GPT-5. Fine-tuned Vintern-1B follows the official recipe: six 448×448 tiles plus a thumbnail per test image and a rank-16 LoRA on the language model for one epoch. The analysis also uses *Global-only*, the bridge of Eq. (1) with $g = 8$ and no local tokens, with and without a decoder LoRA. Our models are *ViBridge-VQA* ($g = 14$, $k = 36$) and *ViBridge-VQA + LoRA*.

4.3 Main Results

ViBridge-VQA nears fine-tuned performance. Without updating any Vintern-1B weight, ViBridge-VQA raises F1 from 17.55 (zero-shot) to 54.56 (Table 2). This is comparable to the reported 53.76 of fine-tuned Vintern-1B despite a stricter split, while the decoder receives 50 rather than 1,792 visual tokens and only 1.35% of the parameters are trained. On the test set, the bridge alone with the same 14 global tokens reaches 51.02 F1 (±0.10), indicating that local tokens add 3.54 F1.

LoRA adds 3.6 F1. With a decoder LoRA, ViBridge-VQA reaches 58.17 F1 (+3.61) while training 1.57% of the parameters, and achieves the best Accuracy (14.67), BLEU (22.53), ROUGE-L (56.25), and METEOR (49.32). Results are stable across seeds (test F1 std of at most 0.19) and splits (test F1 is within 0.4 of validation).

Table 2: Results on AutoViVQA. Baselines are as reported in AutoViVQA [22] and ViMoE-VQA [21]; ours are mean ± std over three seeds on the image-disjoint test split. In bold: the best baseline per column, or our results that exceed it. All values ×100.

Model	Acc	Prec	Rec	F1	BLEU	R-L	MET	CIDEr
Fine-tuned								
ViT5_ViT	7.97	46.84	50.33	48.52	4.13	46.89	31.02	72.68
BARTPhoBEiT	8.81	45.30	46.48	45.88	4.33	44.83	24.57	**188.96**
Vintern-1B	13.01	52.47	55.12	53.76	6.11	51.93	35.25	72.84
ViMoE-VQA	9.55	**62.89**	58.60	**60.69**	12.54	47.12	39.10	88.76
Zero-shot								
Vintern-1B	0.12	17.52	19.87	17.55	1.91	25.84	23.93	8.54
LLaMA 3.2	0.36	23.96	73.71	36.16	3.62	36.11	30.01	62.84
Gemini 2.0 F.	0.55	27.20	74.10	39.79	4.41	39.60	31.72	74.42
Gemini 2.5 F.	0.22	24.43	**76.66**	24.75	0.39	37.27	31.22	71.90
GPT-5	10.84	47.20	55.20	50.89	6.07	47.30	33.34	84.20
Ours ($g = 14$, $k = 36$)								
ViBridge-VQA	11.75 ±.21	55.36 ±.21	56.56 ±.21	54.56 ±.19	**18.52** ±.18	**52.64** ±.19	**45.53** ±.33	110.58 ±.95
ViBridge-VQA + LoRA	**14.67** ±.30	59.02 ±.12	59.96 ±.33	58.17 ±.19	**22.53** ±.11	**56.25** ±.22	**49.32** ±.37	122.88 ±.93

Table 3: Sources of visual tokens with a frozen decoder, on validation. Below each name: number of visual tokens and trained parameters. Best trained model in bold. All values ×100.

Source	Acc	Prec	Rec	F1	BLEU	R-L	MET	CIDEr
Residual 1 token, 4.86M	5.63 ±.59	46.31 ±.34	48.79 ±.33	46.02 ±.36	12.98 ±.15	44.23 ±.38	36.86 ±.31	86.48 ±.77
Global-only 8 tokens, 7.35M	9.21 ±.05	51.54 ±.15	52.48 ±.21	50.74 ±.17	16.88 ±.23	49.03 ±.12	41.38 ±.21	99.07 ±.47
Tile-Attention 8 tokens, 4.14M	4.99 ±1.10	45.49 ±1.10	48.05 ±.60	45.14 ±.92	12.34 ±.32	43.28 ±.95	36.11 ±.73	84.14 ±1.40
Light Q-Former 8 tokens, 27.6M	6.94 ±.32	47.74 ±.80	49.18 ±.51	47.06 ±.66	13.99 ±.33	45.37 ±.63	37.57 ±.60	88.07 ±1.89
Full Q-Former 16 tokens, 69.4M	6.97 ±.33	47.42 ±.70	49.02 ±.35	46.84 ±.47	13.76 ±.36	45.10 ±.41	37.43 ±.34	87.74 ±1.24
C-Abstractor 9 tokens, 19.87M	8.04 ±.07	49.41 ±.32	50.93 ±.32	48.86 ±.33	15.27 ±.13	47.17 ±.33	39.60 ±.33	94.05 ±1.32
Patch-pool 8 tokens, 0.92M	5.58 ±.27	46.11 ±.39	48.52 ±.49	45.78 ±.44	12.71 ±.49	43.99 ±.47	36.69 ±.46	85.83 ±1.43
Local only, untrained 36 tokens, no parameters	0.00	18.07	68.17	27.55	2.47	20.82	34.50	67.20
ViBridge-VQA 50 tokens, 12.86M	**12.38** ±.22	**55.86** ±.21	**56.72** ±.33	**54.96** ±.28	**18.73** ±.23	**53.10** ±.27	**46.01** ±.42	**113.29** ±1.01

4.4 Token Sources

Table 3 compares three sources of visual tokens. *Residual* (one token: a linear projection plus a two-layer MLP correction) and *Global-only* (Eq. (1), $g = 8$)

read the [CLS] vector. Five learned bridges compress the 577 encoder tokens: *Tile-Attention* (self-attention and eight learned queries); the light and full Q-Former [13] (7 and 15 learned queries, respectively, plus an anchor token from the mean encoder feature; two and four layers; the full model also reads the question); C-Abstractor [4] (convolutions around 3×3 pooling; 9 tokens); and *Patch-pool* (a linear projection averaged into eight bands). All are randomly initialised and trained with the same recipe. The third source is our local tokens.

Learned compressors fall short. The five learned compressors stay 1.9–5.6 F1 below Global-only, despite up to 9.4× more parameters. With limited data, a new mapping from patches to the decoder appears hard to learn, and more capacity does not help here: the 69.4M-parameter full Q-Former is no better than the 27.6M light one.

Reusing the pre-aligned projector is more data-efficient. Our results suggest that, under this setup, the same patch features help once they pass through the pre-trained projector. Adding 36 or 144 local tokens to Global-only ($g = 8$) raises F1 by 3.58 and 5.41 (Table 4). At a similar number of tokens, the projector wins with fewer trained parameters: 17 tokens ($g = 8$, $k = 9$) give 52.82 F1 against 46.84 for the 16-token full Q-Former, and 9 tokens ($g = 8$, $k = 1$) give 51.03 against 48.86 for the 9-token C-Abstractor. Pooling alone is not the reason: Patch-pool also averages, but after a new projection, and reaches only 45.78 F1. Distilling the projector into Global-only does not help either: 50.35 F1 with a cosine loss on the mean tokens and 50.46 with a KL loss on the answers, against 50.74 without distillation.

4.5 Token Budget

Local tokens contribute more than global tokens. Table 4 and Fig. 2b vary $g \in \{8, 14\}$ and $k \in \{1, 9, 36, 144\}$. At $g = 14$, going from 1 to 144 local tokens adds 4.94 F1 and 19.05 CIDEr, whereas going from 8 to 14 global tokens adds only 0.2–0.8 F1 at a fixed k. The gain per token falls from 0.07 F1 (23 to 50 tokens) to 0.015 F1 (50 to 158 tokens).

The knee is $g = 14$, $k = 36$. For each of the eight validation metrics, the knee of the Pareto front (Section 3.5, Fig. 2b) is $g = 14$, $k = 36$; the same configuration is selected when generation time is used as the cost. Equal-weight TOPSIS also ranks it first. Although $k = 144$ is 1.65 F1 higher, $k = 36$ captures 72% of its gain over Global-only (67% over the 14-token bridge in Table 5) using 32% of its tokens.

Global tokens peak at $g = 14$. Table 5 and Fig. 2a vary g from 4 to 18 for Global-only. Validation F1 rises from 48.98 ($g = 4$) to 50.74 ($g = 8$) and 51.56 ($g = 14$), then declines to 51.20 and 51.26 at $g = 16$ and 18, respectively; CIDEr shows the same pattern.

Table 4: Effect of the token budget, on validation. Time: minutes to generate all validation answers on one T4. Selected budget in bold. All values ×100.

g	k	Tokens	Time	Acc	Prec	Rec	F1	BLEU	R-L	MET	CIDEr
8	0	8	33.9	9.21 ±.05	51.54 ±.15	52.48 ±.21	50.74 ±.17	16.88 ±.23	49.03 ±.12	41.38 ±.21	99.07 ±.47
14	0	14	37.2	9.74 ±.21	52.47 ±.11	53.16 ±.14	51.56 ±.08	17.76 ±.01	49.86 ±.05	42.14 ±.17	101.18 ±.52
8	1	9	34.2	9.33 ±.29	52.03 ±.12	52.75 ±.10	51.03 ±.12	16.54 ±.27	49.21 ±.13	41.37 ±.11	99.27 ±.58
8	9	17	35.5	10.25 ±.26	53.63 ±.04	54.72 ±.16	52.82 ±.11	17.58 ±.03	50.96 ±.05	43.56 ±.20	106.11 ±.55
8	36	44	36.0	12.15 ±.44	55.19 ±.43	56.14 ±.40	54.32 ±.41	18.22 ±.39	52.48 ±.43	45.31 ±.42	111.88 ±1.43
8	144	152	41.7	13.83 ±.12	56.94 ±.20	58.10 ±.15	56.15 ±.08	19.82 ±.00	54.21 ±.06	47.51 ±.05	119.05 ±.25
14	1	15	35.3	9.67 ±.14	52.50 ±.40	53.48 ±.14	51.67 ±.25	17.30 ±.36	49.93 ±.24	42.36 ±.19	101.66 ±1.11
14	9	23	34.7	10.55 ±.22	53.96 ±.10	54.66 ±.11	52.98 ±.08	17.83 ±.23	51.17 ±.10	43.71 ±.13	106.02 ±.80
14	**36**	**50**	**35.8**	**12.38** ±.22	**55.86** ±.21	**56.72** ±.33	**54.96** ±.28	**18.73** ±.23	**53.10** ±.27	**46.01** ±.42	**113.29** ±1.01
14	144	158	41.1	14.19 ±.36	57.36 ±.07	58.58 ±.03	56.61 ±.04	20.18 ±.13	54.70 ±.05	48.00 ±.05	120.71 ±.12

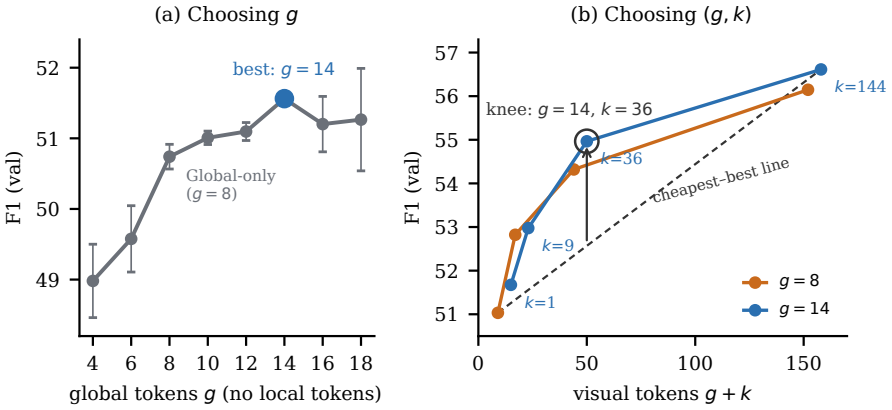


Fig. 2: Budget selection (validation F1). (a) Global-only peaks at $g = 14$. (b) With local tokens, the knee is the configuration furthest above the line joining the cheapest and the most accurate one.

Fewer visual tokens, one encoder pass. Compared with the official tiling, ViBridge-VQA reduces the visual tokens at the decoder input from 1,792 to 50, and the encoder computation from seven 448 × 448 views (362 GMACs each) to one 336 × 336 view (191 GMACs), i.e., 13 times fewer encoder GMACs. Within our budgets, $k = 144$ makes generation about 15% slower than $k = 36$.

Table 5: Global-only with g tokens, on validation. Selected g in bold. All values $\times 100$.

g	Acc	Prec	Rec	F1	BLEU	R-L	MET	CIDEr
4	7.77 ±.27	49.57 ±.48	51.07 ±.57	48.98 ±.52	15.17 ±.48	47.23 ±.58	39.50 ±.59	93.56 ±1.42
6	8.38 ±.39	50.34 ±.53	51.43 ±.43	49.58 ±.47	15.67 ±.29	47.91 ±.48	40.09 ±.50	95.51 ±1.59
8	9.21 ±.05	51.54 ±.15	52.48 ±.21	50.74 ±.17	16.88 ±.23	49.03 ±.12	41.38 ±.21	99.07 ±.47
10	9.24 ±.14	51.95 ±.06	52.58 ±.13	51.01 ±.10	17.03 ±.22	49.26 ±.11	41.46 ±.07	99.20 ±.25
12	9.41 ±.08	52.03 ±.12	52.64 ±.21	51.10 ±.13	17.06 ±.30	49.47 ±.08	41.52 ±.23	99.26 ±.54
14	**9.74** ±.21	**52.47** ±.11	**53.16** ±.14	**51.56** ±.08	**17.76** ±.01	**49.86** ±.05	**42.14** ±.17	**101.18** ±.52
16	9.44 ±.37	51.99 ±.58	52.91 ±.22	51.20 ±.39	17.29 ±.63	49.53 ±.36	41.88 ±.31	100.29 ±.95
18	9.45 ±.45	52.13 ±.72	52.88 ±.69	51.26 ±.73	17.41 ±.54	49.56 ±.68	41.81 ±.69	100.46 ±2.02

Table 6: Effect of a rank-16 decoder LoRA (2.16M parameters, one joint epoch), on test. Bridge parameters: 7.35M for Global-only, 12.86M for ViBridge-VQA. Best in bold. All values $\times 100$.

Model	Acc	Prec	Rec	F1	BLEU	R-L	MET	CIDEr
Global-only	8.86 ±.37	51.19 ±.20	52.37 ±.21	50.48 ±.22	16.24 ±.43	48.70 ±.19	41.10 ±.29	96.20 ±1.10
Global-only + LoRA	10.47 ±.22	53.95 ±.04	54.72 ±.07	53.10 ±.05	19.00 ±.16	51.29 ±.08	43.63 ±.16	104.44 ±.45
ViBridge-VQA	11.75 ±.21	55.36 ±.21	56.56 ±.21	54.56 ±.19	18.52 ±.18	52.64 ±.19	45.53 ±.33	110.58 ±.95
ViBridge-VQA + LoRA	**14.67** ±.30	**59.02** ±.12	**59.96** ±.33	**58.17** ±.19	**22.53** ±.11	**56.25** ±.22	**49.32** ±.37	**122.88** ±.93

4.6 Decoder Adaptation

LoRA gains more with local tokens. ViBridge-VQA + LoRA is the best model on every metric of Table 6 (58.17 F1, 122.88 CIDEr), and LoRA gains more on ViBridge-VQA than on Global-only for all eight metrics (+3.61 versus +2.62 F1, +12.31 versus +8.24 CIDEr). Without local tokens, LoRA lifts Global-only to 53.10 F1, below ViBridge-VQA with a frozen decoder (54.56), and only matches it after three epochs (54.52); Residual, Tile-Attention and the two Q-Formers with LoRA reach 52.69–53.29 validation F1. With a frozen decoder, a trained bridge is needed: local tokens alone give 27.55 F1, mostly because the decoder then answers in long sentences (24.8 words on average).

4.7 Analysis

Local tokens help most on visible content. In Fig. 3, ViBridge-VQA adds 11.0 F1 over Global-only on recognition, 6.4 on action and 5.8 on spatial questions,

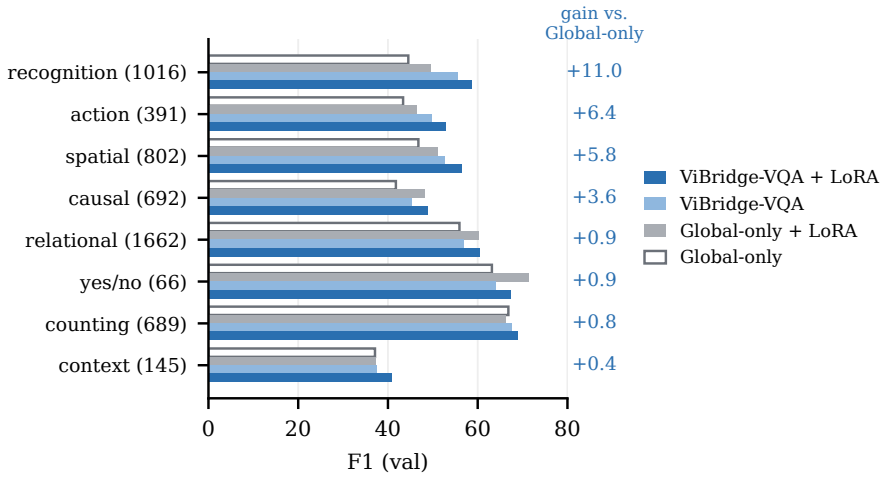


Fig. 3: Validation F1 by reasoning type (number of questions), sorted by the gain of ViBridge-VQA over Global-only (right column).

and beats Global-only + LoRA on these types without adapting the decoder. Decoder adaptation helps more on causal and relational questions, which have the longest reference answers (5.0 and 5.6 words), and on the 66 yes/no questions. ViBridge-VQA + LoRA is the best of the four models on seven of the eight types.

More exact answers, fewer misses. On validation, answers are as long as the references (4.3 words on average), and most of them (77–81%) partially overlap the closest reference. Compared with Global-only, ViBridge-VQA raises exact matches (F1 = 1) from 9.7% to 13.3% and lowers answers with no word in common with any reference (F1 = 0) from 9.6% to 8.0%; with LoRA, these reach 15.7% and 6.8%.

LoRA makes answers more precise. In Fig. 4, ViBridge-VQA already finds the relevant content, such as the driver's phone or the helmets, but its answers are vague. The decoder LoRA makes them precise: the driver is using *the phone*, and the helmets are worn *to protect their heads*.

4.8 Cross-Dataset Results

Table 7 evaluates models trained only on AutoViVQA on ViVQA-X [9] and ViVQA [30], without further training. For each dataset, we draw three random samples of approximately 1,000 test questions (seeds 42, 123, 3407) and evaluate every model on the same questions. Zero-shot Vintern-1B uses six tiles. Even without LoRA, ViBridge-VQA outperforms Global-only + LoRA and zero-shot Vintern-1B (F1: 24.91 versus 15.89 and 14.52 on ViVQA-X; 42.68 versus 31.48 and 23.90 on ViVQA); LoRA further raises F1 to 27.64 and 44.76. Zero-shot Vintern-1B attains the highest Recall only by producing long sentences.



Fig. 4: Validation examples. GT: one of the five references; Ours: ViBridge-VQA; Ours+LoRA: ViBridge-VQA + LoRA.

Table 7: Cross-dataset results (mean ± std over three test samples of about 1,000 questions, 510 for ViVQA; one training seed). Best per dataset in bold. All values ×100.

Model	Acc	Prec	Rec	F1	BLEU	R-L	MET	CIDEr
ViVQA								
Vintern-1B, zero-shot	0.00 ±.00	16.01 ±.18	**60.80** ±1.17	23.90 ±.24	0.27 ±.02	15.78 ±.25	17.84 ±.19	78.63 ±2.07
Global-only + LoRA	7.38 ±.53	31.67 ±.97	34.01 ±1.58	31.48 ±1.15	0.25 ±.09	31.48 ±1.16	21.92 ±.82	84.94 ±2.87
ViBridge-VQA	16.59 ±1.48	40.59 ±1.37	48.86 ±1.89	42.68 ±1.52	0.37 ±.00	42.65 ±1.55	34.62 ±1.37	171.21 ±6.06
ViBridge-VQA + LoRA	**18.93** ±1.37	**43.08** ±.93	50.88 ±1.71	**44.76** ±1.21	**0.63** ±.02	**44.74** ±1.22	**36.98** ±1.28	**183.60** ±7.17
ViVQA-X								
Vintern-1B, zero-shot	0.20 ±.08	9.13 ±.35	**61.15** ±.80	14.52 ±.47	0.06 ±.01	8.98 ±.32	11.96 ±.41	50.13 ±2.83
Global-only + LoRA	2.57 ±.09	13.19 ±.49	26.80 ±1.24	15.89 ±.65	0.18 ±.02	11.94 ±.49	9.08 ±.57	42.12 ±2.92
ViBridge-VQA	7.40 ±.28	20.68 ±.16	41.32 ±.26	24.91 ±.14	**0.34** ±.12	19.38 ±.25	16.00 ±.21	79.51 ±1.15
ViBridge-VQA + LoRA	**8.11** ±.33	**23.07** ±.18	44.57 ±.06	**27.64** ±.11	0.33 ±.05	**21.98** ±.13	**19.24** ±.14	**95.29** ±1.51

5 Conclusion

We introduced **ViBridge-VQA**, which improves Vintern-1B for generative Vietnamese VQA without updating its pre-trained weights. For a VLM whose projector is aligned with its decoder, our results suggest that, in our setup, reusing the projector with a few pooled local tokens is more effective than learning a new connector from limited data. The visual-token budget can be selected with an explicit accuracy–cost criterion, and local tokens complement decoder adaptation: ViBridge-VQA with LoRA achieves the best result.

An existing Vietnamese VLM can be improved by selecting which visual tokens its frozen decoder receives. Code and the image-disjoint split are available at `https://github.com/nguyennn263/modular-vlm-finetune`.

Future work will evaluate ViBridge-VQA on additional datasets and higher-resolution views.

References

1. Alayrac, J.B., Donahue, J., Luc, P., Miech, A., Barr, I., Hasson, Y., Lenc, K., Mensch, A., Millican, K., Reynolds, M., et al.: Flamingo: a visual language model for few-shot learning. In: Advances in Neural Information Processing Systems (NeurIPS). pp. 23716–23736 (2022)
2. Banerjee, S., Lavie, A.: METEOR: An automatic metric for MT evaluation with improved correlation with human judgments. In: Proceedings of the ACL Workshop on Intrinsic and Extrinsic Evaluation Measures for Machine Translation and/or Summarization. pp. 65–72 (2005)
3. Cai, M., Yang, J., Gao, J., Lee, Y.J.: Matryoshka multimodal models. In: Proceedings of the International Conference on Learning Representations (ICLR) (2025)
4. Cha, J., Kang, W., Mun, J., Roh, B.: Honeybee: Locality-enhanced projector for multimodal LLM. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 13817–13827 (2024)
5. Chen, L., Zhao, H., Liu, T., Bai, S., Lin, J., Zhou, C., Chang, B.: An image is worth 1/2 tokens after layer 2: Plug-and-play inference acceleration for large vision-language models. In: Proceedings of the European Conference on Computer Vision (ECCV). Lecture Notes in Computer Science, vol. 15139, pp. 19–35. Springer (2024)
6. Chen, Z., Wang, W., Tian, H., Ye, S., Gao, Z., Cui, E., Tong, W., Hu, K., Luo, J., Ma, Z., et al.: How far are we to GPT-4V? Closing the gap to commercial multimodal models with open-source suites. Science China Information Sciences **67**(12), 220101 (2024)
7. Chen, Z., Wu, J., Wang, W., Su, W., Chen, G., Xing, S., Zhong, M., Zhang, Q., Zhu, X., Lu, L., Li, B., Luo, P., Lu, T., Qiao, Y., Dai, J.: InternVL: Scaling up vision foundation models and aligning for generic visual-linguistic tasks. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 24185–24198 (2024)
8. Doan, K.T., Huynh, B.G., Hoang, D.T., Pham, T.D., Pham, N.H., Nguyen, Q.T.M., Vo, B.Q., Hoang, S.N.: Vintern-1B: An efficient multimodal large language model for Vietnamese (2024), arXiv preprint arXiv:2408.12480
9. Duong, T.B., Tran, H.M., Le-Nguyen, B.N., Duong, D.T.: An automated pipeline for constructing a Vietnamese VQA-NLE dataset. In: Proceedings of the International

Conference on Intelligent Systems and Networks (ICISN). Lecture Notes in Networks and Systems, vol. 1596, pp. 164–173. Springer (2026)
10. Houlsby, N., Giurgiu, A., Jastrzebski, S., Morrone, B., de Laroussilhe, Q., Gesmundo, A., Attariyan, M., Gelly, S.: Parameter-efficient transfer learning for NLP. In: Proceedings of the International Conference on Machine Learning (ICML). pp. 2790–2799 (2019)
11. Hu, E.J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., Chen, W.: LoRA: Low-rank adaptation of large language models. In: Proceedings of the International Conference on Learning Representations (ICLR) (2022)
12. Hwang, C.L., Yoon, K.: Multiple Attribute Decision Making: Methods and Applications. Springer (1981)
13. Li, J., Li, D., Savarese, S., Hoi, S.: BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In: Proceedings of the International Conference on Machine Learning (ICML). pp. 19730–19742 (2023)
14. Li, K.Y., Goyal, S., Semedo, J.D., Kolter, J.Z.: Inference optimal VLMs need fewer visual tokens and more parameters. In: Proceedings of the International Conference on Learning Representations (ICLR) (2025)
15. Li, W., Yuan, Y., Liu, J., Tang, D., Wang, S., Qin, J., Zhu, J., Zhang, L.: Token-Packer: Efficient visual projector for multimodal LLM. International Journal of Computer Vision **133**(10), 6794–6812 (2025)
16. Li, X.L., Liang, P.: Prefix-tuning: Optimizing continuous prompts for generation. In: Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL). pp. 4582–4597 (2021)
17. Lin, C.Y.: ROUGE: A package for automatic evaluation of summaries. In: Proceedings of the ACL Workshop on Text Summarization Branches Out. pp. 74–81 (2004)
18. Lin, T.Y., Maire, M., Belongie, S., Bourdev, L., Girshick, R., Hays, J., Perona, P., Ramanan, D., Zitnick, C.L., Dollár, P.: Microsoft COCO: Common objects in context. In: Proceedings of the European Conference on Computer Vision (ECCV). pp. 740–755 (2014)
19. Liu, H., Li, C., Li, Y., Lee, Y.J.: Improved baselines with visual instruction tuning. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 26296–26306 (2024)
20. Liu, H., Li, C., Wu, Q., Lee, Y.J.: Visual instruction tuning. In: Advances in Neural Information Processing Systems (NeurIPS). pp. 34892–34916 (2023)
21. Nguyen, A.T., Nguyen, T.Q., Dang, D.L., Tran, D.T., Phan, B.D., Nguyen, Q.T., Le, T.: ViMoE-VQA: Reasoning-aware sparse experts for generative Vietnamese visual question answering. Procedia Computer Science (2026), to appear
22. Nguyen, A.T., Phan, B.D., Nguyen, T.Q., Tran, D.T., Dang, D.L., Nguyen, Q.T., Le, T.: AutoViVQA: A large-scale, automatically constructed dataset for Vietnamese visual question answering. In: Proceedings of the Asian Conference on Intelligent Information and Database Systems (ACIIDS). Communications in Computer and Information Science, vol. 2964, pp. 492–506. Springer (2026)
23. Nguyen, N.S., Nguyen, V.S., Le, T.: Advancing Vietnamese visual question answering with transformer and convolutional integration. Computers and Electrical Engineering **119**, 109474 (2024)
24. Nguyen, Q.V., Tran, D.Q., Pham, H.Q., Nguyen, T.K.B., Nguyen, N.H., Nguyen, K.V., Nguyen, N.L.T.: ViTextVQA: A large-scale visual question answering dataset and a novel multimodal feature fusion method for Vietnamese text comprehension in images. Expert Systems with Applications **308**, 130839 (2026)

25. Papineni, K., Roukos, S., Ward, T., Zhu, W.J.: BLEU: a method for automatic evaluation of machine translation. In: Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL). pp. 311–318 (2002)
26. Pham, H.Q., Nguyen, T.K.B., Nguyen, Q.V., Tran, D.Q., Nguyen, N.H., Nguyen, K.V., Nguyen, N.L.T.: ViOCRVQA: Novel benchmark dataset and VisionReader for visual question answering by understanding Vietnamese text in images. Multimedia Systems **31**(2), 106 (2025)
27. Qwen Team: Qwen2.5 technical report (2024), arXiv preprint arXiv:2412.15115
28. Satopää, V., Albrecht, J., Irwin, D., Raghavan, B.: Finding a "kneedle" in a haystack: Detecting knee points in system behavior. In: Proceedings of the International Conference on Distributed Computing Systems Workshops (ICDCSW). pp. 166–171 (2011)
29. Shang, Y., Cai, M., Xu, B., Lee, Y.J., Yan, Y.: LLaVA-PruMerge: Adaptive token reduction for efficient large multimodal models. In: Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV). pp. 22857–22867 (2025)
30. Tran, K.Q., Nguyen, A.T., Le, A.T.H., Nguyen, K.V.: ViVQA: Vietnamese visual question answering. In: Proceedings of the Pacific Asia Conference on Language, Information and Computation (PACLIC). pp. 683–691 (2021)
31. Tran, K.V., Nguyen, K.V., Nguyen, N.L.T.: BARTPhoBEiT: Pre-trained sequence-to-sequence and image transformers models for Vietnamese visual question answering. In: Proceedings of the International Conference on Multimedia Analysis and Pattern Recognition (MAPR). pp. 1–6 (2023)
32. Tran, K.V., Phan, H.P., Nguyen, K.V., Nguyen, N.L.T.: ViCLEVR: A visual reasoning dataset and hybrid multimodal fusion model for visual question answering in Vietnamese. Multimedia Systems **30**(4), 199 (2024)
33. Tsimpoukelli, M., Menick, J., Cabi, S., Eslami, S.M.A., Vinyals, O., Hill, F.: Multimodal few-shot learning with frozen language models. In: Advances in Neural Information Processing Systems (NeurIPS). pp. 200–212 (2021)
34. Vedantam, R., Zitnick, C.L., Parikh, D.: CIDEr: Consensus-based image description evaluation. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR). pp. 4566–4575 (2015)
35. Yang, S., Chen, Y., Tian, Z., Wang, C., Li, J., Yu, B., Jia, J.: VisionZip: Longer is better but not necessary in vision language models. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 19792–19802 (2025)
36. Yao, L., Li, L., Ren, S., Wang, L., Liu, Y., Sun, X., Hou, L.: DeCo: Decoupling token compression from semantic abstraction in multimodal large language models (2024), arXiv preprint arXiv:2405.20985